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The Editorial Board

Neuroinformatics

Springer Nature

Dear Editors,

I am pleased to submit an original research manuscript entitled **Longitudinal Boundary Sharpness Coefficient Slopes Predict Time to Alzheimer Disease Conversion in Mild Cognitive Impairment: A Survival Analysis Using the ADNI Cohort** for consideration in *Neuroinformatics*.

Predicting which individuals with mild cognitive impairment will convert to Alzheimer disease is a problem that sits at the intersection of neuroimaging and clinical informatics. Existing tools either rely on expensive molecular imaging or extract only cross-sectional features from a single scan, missing the dynamic trajectory of tissue change that unfolds across years. This manuscript introduces a fully automated pipeline that derives longitudinal slope features from the Boundary Sharpness Coefficient, a voxel-level index of gray-white matter interface integrity computed from standard T1-weighted MRI. Annualized rates of boundary degradation, estimated by linear regression across four serial scans per subject, are modeled with a Random Survival Forest to produce individual conversion risk scores. Applied to 1,824 scans from 450 ADNI participants, the approach achieved a concordance index of 0.63, a 163 percent improvement over a parametric baseline, using only structural MRI data already collected in routine clinical practice.

Neuroinformatics is a natural home for this work. The manuscript contributes a reproducible, open neuroimaging pipeline, a longitudinal biomarker extraction framework, and a survival modeling approach validated on a large public dataset. These are precisely the kinds of computational tools and datasets that the neuroinformatics community develops and applies. Readers working on brain MRI analysis, longitudinal cohort studies, and data-driven clinical decision support will find the methods and findings directly relevant.

This manuscript is original, has not been published previously, and is not currently under consideration by any other journal. I have no competing interests to disclose. As the sole author, I have approved the final version of the manuscript and take full responsibility for its content.

Thank you for your time and consideration. I look forward to your response.

Sincerely,

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